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stephaniewang.page

Stephanie Wang

Education

Ph.D. and M.S. in Mathematics, UCLA, Eugene V. Cota-Robles Fellow. **2014-2020**
B.S. in Mathematics, National Taiwan University, *magna cum laude*. **2009-2013**

Experience

Research

Research Engineer – reporting to Dr. Qingnan Zhou, *Adobe*, Seattle, WA. **2024-present**
Geometry and physics research and software development, with a focus on geometry processing in machine learning and 3D modeling.

Postdoc – with Prof. Albert Chern, *UCSD*, San Diego, CA. **2020-2023**
Geometric deep learning, geometry optimization and physics-based simulation of various materials (solids, fluids, waves, cloth, hair, frictional contact, plasticity, fracture, *etc.*). Mentored students: [Mohammad Sina Nabizadeh](#), [Shiyang Jia](#), [Chad McKell](#), [Hang Yin](#), [Baichuan Wu](#).

Ph.D. Study – with Prof. Wilfrid Gangbo, *UCLA*, Los Angeles, CA. **2019-2020**
Regularity theory for minimizers of polyconvex functionals related to Navier-Stokes equation.

Exchange Study – with Prof. Johan Gaume, *EPFL*, Lausanne, Switzerland. **2019 summer**
Simulating snow and tire interaction using physics-based simulation, in collaboration with Michelin tires.

Ph.D Study – with Prof. Joseph Teran, *UCLA*, Los Angeles, CA. **2016-2019**
Physics-based simulation of various materials (fracture, cloth, hair, deformable body, *etc.*) using Material Point Method and Finite Element Method.

Tech Intern, *Walt Disney Animation Studio*, Burbank, CA. **2018 summer**
R&D for pioneering simulation technology in animated feature films, teaming with VFX artists.

Teaching

Assistant Adjunct Professor / Instructor, *UCLA Math Dept*, Los Angeles, CA. **2019-2020**
Taught three courses: linear algebra, machine learning (remote) and multivariable calculus (remote).

Teaching Assistant, *UCLA Math Dept*, Los Angeles, CA. **2015-2020**
TA-ed courses: linear algebra, numerical methods (intro, intermediate, and advanced), C++ programming (intro, intermediate, and advanced).

Skills

Programming: C++, Python, MATLAB, $\LaTeX$, Vim, git, zsh, Houdini

Math: Optimization, differential geometry, solid and fluid dynamics, scientific computing

Languages: English and Mandarin Chinese (bilingual)

Technical communication: 12 papers in top journals and 18 talks at top conferences / institutes.

Selected Publications

[Implicit UVs: Real-time Semi-global Parameterization of Implicit Surfaces](#). Computer Graphics Forum (Eurographics 2025).

Baptiste Genest, Pierre Gueth, Jérémy Levallois, [Stephanie Wang](#).

[Exterior Calculus in Graphics: Course Notes for a SIGGRAPH 2023 Course](#). SIGGRAPH 2023. [Stephanie Wang](#), [Mohammad Sina Nabizadeh](#), [Albert Chern](#).

[DeepCurrents: Learning implicit representations of shapes with boundaries](#). CVPR 2022.

[David Palmer](#), [Dmitriy Smirnov](#), [Stephanie Wang](#), [Albert Chern](#), and [Justin Solomon](#).

[Computing minimal surfaces with differential forms](#). ACM ToG (SIGGRAPH 2021). [Stephanie Wang](#) and [Albert Chern](#).

[Simulation and visualization of ductile fracture with the material point method](#). PACM-CGIT (SCA 2019 **Best Paper Award**).

[Stephanie Wang](#), [Mengyuan Ding](#), [Theodore F. Gast](#), [Leyi Zhu](#), [Steven Gagniere](#), [Chenfanfu Jiang](#), and [Joseph M. Teran](#).

...and 7 more papers published in top journals in computer graphics and other scientific fields.